DAX & Columns Functions

DAX Functions

This document provides a summary of various DAX (Data Analysis Expressions) measures used in a cricket analysis dashboard. Each measure includes its purpose and the DAX formula used for calculation.

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1. Total Runs

- **Purpose**: Total number of runs scored by the batsman.
- **DAX Formula**:

  `Total Runs = SUM(fact_batting_summary[runs])`

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2. Total Innings Batted

- **Purpose**: Total number of innings a batsman got a chance to bat.
- **DAX Formula**:

  `Total Innings Batted = COUNT(fact_batting_summary[match_id])`

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3. Total Innings Dismissed

- **Purpose**: Total number of innings batsman got out.
- **DAX Formula**:

  `Total Innings Dismissed = SUM(fact_batting_summary[out])`

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4. Batting Average

- **Purpose**: Average runs scored in an innings.
- **DAX Formula**:

  `Batting Avg = DIVIDE([Total Runs], [Total Innings Dismissed], 0)`

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5. Total Balls Faced

- **Purpose**: Total number of balls faced by the batsman.
- **DAX Formula**:

  `total balls faced = SUM(fact_batting_summary[balls])`

6. Strike Rate

- **Purpose**: Number of runs scored per 100 balls.
- **DAX Formula**:

  `Strike rate = DIVIDE([Total Runs], [total balls faced], 0) * 100`

7. Batting Position

- **Purpose**: Batting position of a player.
- **DAX Formula**:

  `ROUNDUP(AVERAGE(fact_batting_summary[batting_pos]), 0)`

8. Boundary %

- **Purpose**: Percentage of boundaries scored by the batsman.
- **DAX Formula**:

  `DIVIDE(SUM(fact_batting_summary[Boundary runs]), [Total Runs], 0)`

9. Average Balls Faced

- **Purpose**: Average balls faced by the batter in an innings.
- **DAX Formula**:

  `AVERAGE(fact_batting_summary[balls])`

10. Wickets

- **Purpose**: Total number of wickets taken by a bowler.
- **DAX Formula**:

  `wickets = SUM(fact_bowling_summary[wickets])`

11. Balls Bowled

- **Purpose**: Total number of balls bowled by the bowler.
- **DAX Formula**:

  `balls bowled = SUM(fact_bowling_summary[balls])`

12. Runs Conceded

- **Purpose**: Total runs conceded by the bowler.
- **DAX Formula**:

  `Runs Conceded = SUM(fact_bowling_summary[runs])`

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13. Bowling Economy

- **Purpose**: Average number of runs conceded in an over.
- **DAX Formula**:

  `Economy = DIVIDE([Runs Conceded], ([balls Bowled] / 6), 0)`

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14. Bowling Strike Rate

- **Purpose**: Number of balls bowled per wicket.
- **DAX Formula**:

  `Bowling Strike Rate = DIVIDE([balls Bowled], [wickets], 0)`

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15. Bowling Average

- **Purpose**: Number of runs allowed per wicket.
- **DAX Formula**:

  `Bowling Average = DIVIDE([Runs Conceded], [wickets], 0)`

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16. Total Innings Bowled

- **Purpose**: Total number of innings bowled by a bowler.
- **DAX Formula**:

  `Total Innings Bowled = DISTINCTCOUNT(fact_bowling_summary[match_id])`

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17. Dot Ball %

- **Purpose**: Percentage of dot balls bowled by a bowler.
- **DAX Formula**:

  `Dot Ball % = DIVIDE(SUM(fact_bowling_summary[zeros]), SUM(fact_bowling_summary[balls]), 0)`

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18. Player Selection

- **Purpose**: To understand if a player is selected or not.

- **DAX Formula**:

```
Player Selection =
IF(ISFILTERED(dim_player[name]), "1", "0")
```

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19. Display Text

- **Purpose**: To display a text if no player is selected.
- **DAX Formula**:

```
Display Text =
if([Player Selection] = "1", "",
"Select Player(s) by clicking the player's name to see their individual or combined strength.")
```

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20. Color Callout Value

- **Purpose**: To display a value only when a player is selected.
- **DAX Formula**:
  `Color Callout Value = if([Player Selection] = "0", "#D8CF1D", "#1D1D2E")`

Calculated Columns

These calculated columns are used to derive specific insights directly within the data model, using DAX expressions.

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1. Boundary Runs

- **Purpose**: To find the total number of runs scored by hitting fours and sixes.
- **DAX Formula**:
  `boundary runs = fact_batting_summary[fours]*4 + fact_batting_summary[sixes]*6`
- **Table**: `fact_batting_summary`

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2. Boundary Runs Bowling

- **Purpose**: To find the total number of runs conceded by bowlers in boundaries.
- **DAX Formula**:
  `Boundary runs = fact_bowling_summary[fours]*4 + fact_bowling_summary[sixes]*6`
- **Table**: `fact_bowling_summary`

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3. Custom Batting Order

- **Purpose**: To assign the batting order to a potential final 11.
- **DAX Formula**:

```
DAX
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SWITCH(
    TRUE(),
    dim_player[name] = "Jos Buttler", 1,
    dim_player[name] = "Rilee Rossouw", 2,
    dim_player[name] = "Alex Hales", 2,
    dim_player[name] = "Virat Kohli", 3,
    dim_player[name] = "Suryakumar Yadav", 4,
    dim_player[name] = "Glenn Phillips", 5,
    dim_player[name] = "Marcus Stoinis", 6,
    dim_player[name] = "Glenn Maxwell", 6,
    dim_player[name] = "Sikandar Raza", 7,
    dim_player[name] = "Rashid Khan", 8,
    dim_player[name] = "Shadab Khan", 8,
    dim_player[name] = "Sam Curran", 9,
    dim_player[name] = "Shaheen Shah Afridi", 10,
    dim_player[name] = "Anrich Nortje", 11
)
```

- **Table**: `dim_player`